[INCH-POUND]
A-A-2678A
August 10, 1995
SUPERSEDING
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July 1, 1991

COMMERCIAL ITEM DESCRIPTION

WRENCH, IMPACT, PNEUMATIC, PORTABLE (PISTOL GRIP)

This Commercial Item Description is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. Scope. This commercial item description covers portable, pneumatic impact wrenches, used for tightening and loosening bolts and nuts.

2. Salient Characteristics.

2.1 Design. The wrench shall be similar to Figure 1 and comply with the performance criteria in Table I. The wrench shall be equipped with a pistol-grip handle and shall operate properly at an air pressure of 80 to 100 psig.

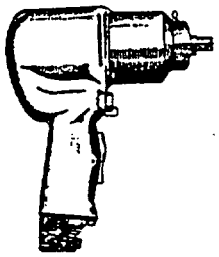


Figure 1. Portable pneumatic impact wrench.

Table I

National Stock Number	Spindle Square Drive Size (Inch)	Spindle Offset Maximum (Inch)	Air Inlet NPT NPTF (Inch)	Air Consumption While Impacting CFM (Maximum)	Torque Developed After 5 Seconds Impacting Minimum (ft.-lbs.)
5130-00-234-4879	3/8	1-3/8	1/4	15	70
5130-00-889-2134	1/2	1-3/8	1/4	30	250
5130-00-184-1427	3/4	2	3/8	50	750
5130-01-411-3394	1	2-1/2	3/8	60	1100

2.2 Material: The material used in the construction of the wrenches shall be of good commercial quality suitable for the intended purpose. All materials used shall be free from defects and imperfections that may adversely affect the serviceability of the finished product. A metal or composite housing is required. Cadmium is not an acceptable plating on any component.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any other data which may improve this document should be sent to: General Services Administration, Federal Supply Service, Tools and Appliances Commodity Center (6FETE-CO), Washington, DC 20406.

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2.3 Lubrication: Provisions shall be made for lubrication to be supplied to moving parts in the path of the air stream. The oil may be applied manually into the motor or supplied automatically through the air supply line. Other suitable means shall be provided for supplying lubrication to moving parts not in the air stream.

2.4 Air inlet connection: The air inlet connection shall consist of a well-secured ferrous fitting threaded with internal American Standard taper pipe threads (NPT) or internal Dryseal American Standard taper pipe threads (NPTF) as specified in Table I. The air inlet shall be fitted with an air strainer or screen (20 mesh or finer) which shall be effective in preventing solid particles in the compressed air supply from entering the tool. The strainer or screen shall be of such design that it can be easily removed for cleaning.

2.5 Bearings: The rotor shaft shall be provided with ball bearings. Needle and roller bearings, which are primarily designed for radial loads, shall not be used in a thrust position. If open type bearings are used, they shall be suitably housed and adequately sealed to prevent leakage of lubricant and any entrance of dirt and dust. Sleeve bearings are not acceptable.

2.6 Spindle: The drive end of the spindle shall be wear-resistant. The drive end, when fitted with a socket, shall not show excessive run-out which may adversely affect operation of the wrench. The drive end of the spindle shall conform to ANSI B107.4, spring loaded plunger or through hole design.

2.7 Operation. The wrenches shall be reversible. The rotational direction of the wrench shall be controlled by a conveniently located mechanical device which is independent of the throttle. The rotational control shall be plainly and permanently marked to indicate direction of rotation.

2.8 Impact unit: Wrenches shall contain an impact unit which is actuated by the motor. The impact unit shall be designed to convert torque from the motor into rotary impacts and to transmit the impacts to the spindle.

2.9 Throttle: Wrenches shall be provided with a nonleakable manual throttle for controlling the supply of compressed air. The wrench shall be equipped with a trigger or button style throttle.

2.10 Air volume (torque) regulator: The wrenches shall be furnished with an air volume (torque) regulator. The regulator shall be built into the tool. The regulator shall have varying degrees of restriction for regulating the volume of air flow. When in the wide open position, the wrench air consumption shall not exceed that shown in Table I.

2.11 Manual: One copy of a technical manual consisting of the manufacturer's standard commercial instruction and parts list shall be furnished with each tool.

2.12 Workmanship: Details of workmanship shall be in accordance with the best commercial standards and practices. Paints, coatings, platings, and finishes shall be smooth, dry, adherent, continuous, and not stained or discolored. Fasteners shall be firmly secured and show no evidence of deformation, cross threading, or hazardous burrs. Adhesives and lubricants adequate for their intended purpose shall be properly and neatly applied. Adhesives shall be adequately cured. Wires and cables shall be neatly dressed and shall not be frayed or in contact with sharp edges. There shall be no interference, binding, or galling of parts. External and bearing surfaces shall be free of tool and gouge marks, nicks, or other surface imperfections. The item shall be clean and free of corrosion and debris (e.g., chips, shavings, slivers) or other foreign material. The item shall be free from manufacturing workmanship defects (e.g., loose, missing, binding or misaligned parts, sharp or rough external edges, corners or surfaces) and material workmanship defects (e.g., pits, rips, fins, burrs, tears, nodules, cracks, blisters) which may adversely impact the item's serviceability, durability, safety, or appearance.

3. Regulatory Requirements: In accordance with Section 23.403 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing supplier's employees to undue hazards from the recovered materials.

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3.1 Metric products: Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the specification preparing activity for change to this document.

4. Quality assurance provisions.

4.1 Responsibility for inspection: Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services adhere to prescribed requirements.

4.2 Inspection: It shall be determined that the sample units meet all salient characteristics stated in the requirements section of this document. Any sample containing one or more defects shall be rejected, and if the number of rejected units exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.3 Test conditions: All tests shall be conducted with an air supply of 90 psig (620 kPa) measured as close to the tool as practicable. The test tool shall be connected to the air supply line by means of a 10 foot length of air hose having the following hose size (inner diameter): 5/16 inch hose size for wrenches with a 3/8 inch drive size; 3/8 inch hose size for wrenches with a 1/2 inch drive size; 1/2 inch hose size for wrenches with a 3/4 inch drive size; 3/4 inch hose size for wrenches with a 1 inch drive size. The air supply line shall be equipped with a filter, air pressure regulator and lubricator.

4.4 First article inspection. When specified, the contractor shall furnish one sample of a completely fabricated tool for first article inspection. The sample shall have been manufactured and inspected using the same materials, processes, and methods that will be used in the production of the tools used to fulfill the terms of the contract or purchase order. The sample shall meet the requirements of this Commercial Item Description. Approval of the first article sample shall not relieve the contractor of the responsibility to furnish tools in accordance with the requirements of this Commercial Item Description. First article inspection shall consist of the following evaluation in the sequence listed: Inspection, Air consumption test,Torque test, Sound pressure level test, Practical service test, Service test, and Examination after service test.

4.5 Production lot inspection and testing: Production lot testing shall consist of the following evaluations conducted in the sequence listed: Inspection, Air consumption test, Torque test, and Practical service test. A sample lot shall consist of all items of same NSN offered for delivery at one time. Sampling procedures shall be in accordance with ANSI/ASQC Z 1.4 at an Inspection Level of S-3. The acceptable quality level (AQL) shall be 1.5 defects per 100 units tested. A sample unit shall consist of one completely fabricated tool.

4.6 Air consumption test: Air consumption of the tool shall be measured after 5 seconds of impacting. Air consumption shall be measured with either a certified liquid displacement meter or a flowmeter.

4.7 Torque test: The torque test shall be conducted on a torque analyzer of appropriate capacity calibrated to assure accurate readings.

4.8 Sound pressure level test: The sound pressure level test shall be conducted in accordance with the applicable provisions of the Compressed Air and Gas Institute's, "CAGI-PNEUROP Test Code for the Measurement of Sound from Pneumatic Equipment". The sound pressure level shall not exceed 93 decibels for size 3/8 inch, 95 decibels for size 1/2 inch and 3/4 inch, and 103 decibels for the 1 inch size impact wrench.

4.9 Practical service test: Each sample tools shall be operated for 5 minutes. The test shall duplicate a typical work application. During the test, the tool shall be checked for throttle operation, rotation, excessive noise or vibration, ease of operation, and general serviceability.

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4.10 Service test: A simulated service test shall be conducted for 150 hours. The test sample shall be installed in a fixture in a manner that the tool can be loaded and automatically actuated for 150 hours. Each cycle shall be 10 seconds ON and 20 seconds OFF.

4.11 Examination after service test: Tools satisfactorily completing the service test shall be subjected to the air consumption test, torque test, and sound pressure level test. The test sequence shall be as specified above. Upon completion of these tests, the tool shall be disassembled and the parts examined. Any tool having broken, cracked, or deformed parts shall be rejected.

5. Preservation, packaging, packing, and marking: Tools shall be preserved, packaged, packed, and marked as specified in the contract or purchase order.

5.1 Identification markings: Each tool (including all items within a set or kit) shall be marked with the manufacturer's name or identifying symbol, and the state or country of manufacture, unless otherwise specified. All marking shall be engraved, etched, molded, or indented directly on the item's surface in such a manner that it remains clearly legible throughout the life of the item.

6. Notes:

6.1 The issue of referenced documents in effect on the date of invitation for bids or request for proposal shall be used to determine compliance with the stated requirements. Sources for obtaining referenced documents may be determined by contacting the Contracting Officer.

6.2 Military standards are available from Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19120-5094.

6.3 ANSI standards are available from the American National Standards Institute, 1430 Broadway, New York, NY 10018.

6.4 The "CAGI-PNEUROP Test Code for the Measurement of Sound from Pneumatic Equipment", is available from the Compressed Air and Gas Institute, 1300 Sumner Avenue, Cleveland, OH 44115.

Military Coordinating Activity

Army - GL

Preparing Activity

GSA - FSS

Custodians

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Review Activities

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